

Asymptotic expansion of the variation of the Quillen metric and its moment map interpretation

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- In this talk, I will mainly talk about extension of Donaldson-Fujiki moment map interpretation of scalar curvature to other scalar curvature quantities on Kähler metrics.
- Recently Dervan-Hallam propose a general framework for such problem under the name of *Z-critical equations*.
- However, our viewpoint is related, but different from theirs.
- I will also discuss about generalization of Mabuchi functional and Futaki invariant for these quantities.

Moment map

- Let (M, Ω) be a symplectic manifold, G a Lie group acting on M preserving Ω .
- Let $\mathfrak{g} := \text{Lie}(G)$ and $\widehat{X}$ for $X \in \mathfrak{g}$ denotes the fundamental vector field on M generated by X .

Definition

An equivariant map $\mu : M \rightarrow \mathfrak{g}^*$ is called *moment map* if

$$d(x \mapsto \langle \mu_x, X \rangle) = \iota_{\widehat{X}} \Omega$$

for $\forall g \in G, \forall x \in M, \forall X \in \mathfrak{g}$.

Recall on the Donaldson-Fujiki picture

- Let (M, ω) be a symplectic manifold. Let $\mathcal{J}_{int}$ be the space of integrable compatible almost complex structures on M . That is, for each $J \in \mathcal{J}_{int}$ corresponds to a Kähler manifold $(M, \omega, g_{(\omega, J)}, J)$.
- The set $\mathcal{J}_{int}$ has a natural structure of infinite-dimensional Kähler manifold with Kähler form Ω_{DF} .
- The group of Hamiltonian diffeomorphisms $\mathcal{G} := \text{Ham}(M, \omega)$ acts on $\mathcal{J}_{int}$ preserving Ω_{DF} .

Donaldson-Fujiki '97, '92

The map $\mu : \mathcal{J}_{int} \rightarrow C^\infty(M) \simeq \text{Lie}(\mathcal{G})^*$ defined by

$$\mu(J) := \text{Scal}(g_{(\omega, J)})$$

is a moment map for $\mathcal{G} \curvearrowright (\mathcal{J}_{int}, \Omega_{DF})$.

- It is a basis of YTD correspondence with analogy with Kempf-Ness theorem.
- The original proof of Donaldson-Fujiki is by direct computation using variation formula of scalar curvature, and is difficult to generalize this approach to more complicated curvature quantities.
- Dervan-Hallam derived the moment map property of wide variety of scalar curvature quantities called Z-critical equations by the language of equivariant differential forms.
- In the context of Kempf-Ness theorem, it is natural to derive moment map from $\mathcal{G}$ -equivariant line bundle over $\mathcal{J}_{int}$ and metric on it.
- In this talk, we will use $\mathcal{G}$ -equivariant determinant line bundle and Quillen metric to generalize Donaldson-Fujiki picture.

Statement/idea of the result

- Assume $[\omega]$ is integral and fix a prequantization $P = (L, h, \nabla)$. That is, complex line bundle L on M , hermitian metric h on L , unitary connection ∇ such that $\nabla^2 = -i\omega$.
- Using L , we can define a $\mathcal{G}$ -equivariant line bundle λ on $\mathcal{J}_{int}$. The curvature form of λ yields $\mathcal{G}$ -invariant 2-form Ω and corresponding moment map $\mu : \mathcal{J}_{int} \rightarrow C^\infty(M)$.
- Now repeat this with L replaced by $L^k, k \in \mathbb{N}$. For each k , we get $\mathcal{G}$ -invariant 2-form $\Omega^{(k)}$ and corresponding moment map $\mu^{(k)} : \mathcal{J}_{int} \rightarrow C^\infty(M)$.

- For each k , we have the moment map equation

$$\iota_{\hat{f}}\Omega^{(k)} = d\langle f, \mu^{(k)} \rangle_{L^2}.$$

- As $k \rightarrow \infty$, $\Omega^{(k)}$ and $\mu^{(k)}$ admits an asymptotic expansion:

$$\Omega^{(k)} = \sum_{j=0}^{\infty} k^{n+1-j} \Omega_j, \quad \mu^{(k)} = \sum_{j=0}^{\infty} k^{n+1-j} \mu_j.$$

- As this asymptotic expansion commutes with derivatives, taking $k \rightarrow \infty$ of the moment map equation, we get

$$\iota_{\hat{f}}\Omega_j = d\langle f, \mu_j \rangle_{L^2}$$

for each $j \geq 0$.

- First few expressions of μ_j :

$$\mu_0 = 1;$$

$$\mu_1 = \frac{1}{2}S;$$

$$\mu_2 = -\frac{1}{6}\Delta S + \frac{1}{24}(|R|^2 - 4|Ric|^2 + 3S^2).$$

- First few expressions of Ω_j (upto constant):

$$\Omega_0 = 0;$$

$$\Omega_1 = \int_M \text{Tr}(J \cdot \cdot) \frac{\omega^n}{n!};$$

$$\Omega_2 = \int_M \text{Tr}(J \cdot \cdot) S(g_{(\omega, J)}) \frac{\omega^n}{n!} + \int_M s_1(\cdot, \cdot) \frac{\omega^n}{n!}.$$

- Since $\Omega_1 = \frac{1}{2}\Omega_{DF}$, we recover Donaldson-Fujiki picture by taking $j = 1$.

- In finite dimensional setting of Kempf-Ness theorem, the moment map can be computed by taking derivative of log-norm in the direction of complexified orbit.
- In our setting $\mathcal{G} \curvearrowright \mathcal{J}_{int}$ however, there is no genuine complexification of $\mathcal{G}$.
- Instead, one can *regard* the complexified orbit of $\mathcal{G} \curvearrowright \mathcal{J}_{int}$ as fixing J and changing the Kähler form ω inside of the de Rham class $[\omega]$. i.e. changing the Kähler potentials.
- Then indeed, the moment maps μ_j appear in the following way: fix J and take the derivative of the log of the *Quillen metric* $|| \cdot ||$ on $\lambda^{(k)}$. It has an asymptotic expansion as $k \rightarrow \infty$:

$$\frac{\delta}{\delta \varphi} \log || \cdot ||_{\omega_{\varphi}, h_{\varphi}^k}^2 = \sum_{j=0}^{n+1} k^{n+1-j} \langle \delta \varphi, \mu_j \rangle_{L^2}.$$

Fix a complex structure J and take the derivative of the logarithm of the Quillen metric with respect to the Kähler potential φ . Its asymptotic expansion as $k \rightarrow \infty$ has the following form:

$$\frac{\delta}{\delta\varphi} \log || \cdot ||_{\omega_\varphi, h^k e^{-k\varphi}}^2 = \sum_{j=0}^{n+1} k^{n+1-j} \int_M \delta\varphi \mu_j(g_{(\omega_\varphi, J)}) \frac{\omega_\varphi^n}{n!}$$

for some functions $\mu_j \in C^\infty(M)$ depending on the curvature of the Kähler metric $g_{(\omega_\varphi, J)}$. Then the map $\mu_j : \mathcal{J}_{int} \rightarrow C^\infty(M)$ defined by

$$\mu_j(J') := \mu_j(g_{(\omega, J')})$$

is a moment map for $\mathcal{G} \curvearrowright (\mathcal{J}_{int}, \Omega_j)$, where $\Omega_j \in \mathcal{A}^2(\mathcal{J}_{int})$ is given by the following fiber integral

$$\Omega_j = \int_{\mathcal{X}/\mathcal{J}_{int}} \text{Td}_j(TM, g_{(\omega, J')}) \text{ch}_{n+1-j}(\tilde{L}, \tilde{h}).$$

- It also provides log-norm functionals ($\approx$ Mabuchi functional) and weights ($\approx$ Futaki invariants).
- Hence asymptotically, this fits well in the standard formal GIT picture, except for the positivity of Ω_j . This is necessary for the convexity of the log-norm functional.
- We do not expect it in general.
- Q) Under what assumption it is positive and (infinite dimensional) Kempf-Ness theorem such as YTD correspondence can be formulated?, or, is it a natural question to ask?

Space of compatible almost complex structures

- Fix (M, ω) , and let $\mathcal{J}$ be the set of *compatible almost complex structures* on M , i.e.

$$\mathcal{J} = \{J \in \Gamma(M, \text{End} TM) : g_{(\omega, J)}(\cdot, \cdot) := \omega(\cdot, J\cdot) \text{ is positive}\}.$$

Thus for $J \in \mathcal{J}$, $g_{(\omega, J)}$ is an associated Riemannian metric.

- There is a natural (almost) complex structure $\mathbb{J}$ on $\mathcal{J}$ by

$$\begin{aligned}\mathbb{J}_J : T_J \mathcal{J} &\rightarrow T_J \mathcal{J} \\ A &\mapsto JA.\end{aligned}$$

- Let $\mathcal{J}_{int}$ be a complex subspace of integrable J 's.

- The group $\text{Ham}(M, \omega)$ of Hamiltonian diffeomorphisms of M acts (holomorphically) on $\mathcal{J}$ by pull-back.
- We can form a *universal family* of Kähler manifolds

$$\pi : \mathcal{X} = \bigcup_J M_J \rightarrow \mathcal{J}_{int}$$

where $M_J := (M, J)$ is the complex manifold corresponding to J .

- Assume $[\omega]$ is integral and fix a prequantization $P = (L, h, \nabla)$ of (M, ω) . For each $J \in \mathcal{J}_{int}$, the pair J and ∇ uniquely determines a holomorphic structure on L . Denote this holomorphic line bundle by L_J .
- These define a hermitian holomorphic line bundle $(\tilde{L}, \tilde{h})$ on $\mathcal{X}$.

Determinant line bundle and Quillen metric

- Let $\pi : \mathcal{X} \rightarrow B$ be a proper holomorphic submersion of relative dimension n with connected fibers $M_y := \pi^{-1}(y)$ with smooth family of Kähler metrics g^M on fibers. Assume the fibration is *Kähler*.
- Let ξ be a hermitian holomorphic vector bundle on $\mathcal{X}$. Construction of Bismut–Gillet–Soule gives a holomorphic line bundle called *determinant line bundle* $\lambda(\xi)$ on B with fiber

$$\lambda_b = \bigotimes_{p=0}^n (\det H^p(M_b, \xi|_{M_b}))^{(-1)^{p+1}}.$$

- To define a smooth hermitian metric on $\lambda(\xi)$, multiply natural L^2 metric with *Ray-Singer analytic torsion*, to get a *Quillen metric*:

$$|||_b = ||_b \tau_b(\bar{\partial});$$

where $||$ denotes L^2 metric for Harmonic representatives, and $\tau_b(\bar{\partial})$ denotes the Ray-Singer analytic torsion defined by

$$\tau_b(\bar{\partial}) = \prod_p \det'(\square_\xi|_{\Omega^{0,p}})^{p(-1)^{p+1}/2}.$$

- It defines a Chern connection ${}^1\nabla$ on $\lambda(\xi)$. There are two important Theorems about Quillen metric, which we state in the next two slides.

1. Quillen (BGS) anomaly formula

By construction, Quillen metric depends on fiberwise Kähler metric and hermitian metric h on ξ . For $B = \{point\}$, if we change the metrics, we have a following.

Quillen anomaly formula (Bismut-Gillet-Soulé '88)

$$\log \frac{\|\cdot\|_{g',h'}^2}{\|\cdot\|_{g,h}^2} = i \int_M BC(\text{Td}; g', g) \text{ch}(\xi, h) + \text{Td}(TM, g') BC(\text{ch}; h', h).$$

Here BC denotes the *Bott-Chern form*. It is a holomorphic version of transgression form:

$$\bar{\partial} \partial BC(\phi; h_1, h_0) = \phi(\text{curv}(h_1)) - \phi(\text{curv}(h_0)).$$

2. BGS curvature formula

Let ${}^1\nabla$ be a Chern connection on $\lambda(\xi)$ with respect to the Quillen metric $|| \cdot ||$. Then its curvature is given by the following.

BGS curvature formula (Bismut-Gillet-Soulé '88)

The curvature of ${}^1\nabla$ is given by

$$({}^1\nabla)^2 = i \left[\int_{\mathcal{X}/B} \text{Td}(TM, g^M) \text{ch}(\xi, h^\xi) \right]^{(2)}$$

where $[\]^{(2)}$ denotes the component of degree 2.

Variation of the Quillen metric

- Here we fix J and vary $\varphi \in \mathcal{K}_\omega$ in the space of Kähler potentials.
- Let (M, ω) be a polarized Kähler manifold with polarizing line bundle L .
- For each Kähler metric $\omega_\varphi \in [\omega]$, there exists a Hermitian metric $h_\varphi = h e^{-\varphi}$ on L such that $(\nabla^{h_\varphi})^2 = -i\omega_\varphi$.
- Let $\lambda(L^k)$ be the determinant line with $\xi = L^k$, $B = \{\text{point}\}$.
- Let $|||_{\omega_\varphi, h^k e^{-k\varphi}}$ be the Quillen metrics constructed by $g^M = g_{\omega_\varphi}$, $h^\xi = h^k e^{-k\varphi}$.
- i.e. Kähler potential is all of the metric data.

- We compute the asymptotic expansion of $\log \frac{|||_{\omega, h^k}^2}{|||_{\omega_\varphi, h^k e^{-k\varphi}}^2}$ as $k \rightarrow \infty$ by two different methods.
- First method is obviously to use the Quillen anomaly formula and expand it in k :

Method 1

$$2 \log \frac{|||_{\omega, h^k}}{|||_{\omega_\varphi, h^k e^{-k\varphi}}} = \sum_{j=0}^{n+1} k^{n+1-j} \int_M -iBC(\text{Td}_j; g_\varphi, g) \frac{\omega^{n+1-j}}{(n+1-j)!} \\ + \text{Td}_j(R_\varphi) \frac{-1}{(n+1-j)!} \sum_{s=0}^{n-j} \varphi \omega_\varphi^s \wedge \omega^{n-j-s}.$$

- Second method uses TYZ expansion of Bergman kernel and recently proved asymptotic expansion of the Ray-Singer torsion.
- By construction, one can obtain the asymptotic expansion of the Quillen metric from asymptotic expansions of (1) the L^2 -product on $\det H^0(M, L^k)$ and (2) the Ray-Singer analytic torsion $\tau_{(\omega, h^k)} := \tau_{(\omega, h^k)}(\bar{\partial}^{(0, \infty)})$.
- The asymptotic expansions of the L^2 -products can be obtained from Donaldson's lemma and TYZ expansion of the Bergman kernel:

$$\rho_k(\omega) = \sum_{j=0}^{\infty} k^{n-j} a_j(g_\omega)$$

where $\rho_k(\omega)$ is the diagonal of the k^{th} Bergman kernel associated with (ω, h^k) .

- For example, $a_0 = 1$; $a_1 = \frac{1}{2}S$; $a_2 = \frac{1}{3}\Delta S + \frac{1}{24}(|R|^2 - 4|Ric|^2 + 3S^2)$.

- For the Ray-Singer analytic torsion $\tau_{(\omega, h^k)}$, Finski recently established the full asymptotic expansion generalizing the result of Bismut-Vasserot:

Finski '18

There are local coefficients $\alpha_j, \beta_j \in \mathbb{R}, j \in \mathbb{N}$ such that as $k \rightarrow \infty$,

$$-2 \log \tau_{(\omega, h^k)} = \sum_{j=0}^{\infty} k^{n-j} (\alpha_j \log k + \beta_j(g_{\omega})). \quad (0.1)$$

The coefficients α_j do not depend on ω . Moreover, derivatives over the base commute with the asymptotics (0.1).

- Here, the local coefficient means that it can be expressed as an integral of a density defined locally over M , given by g_{ω} and its derivatives.

- Combining these, we obtain

Method 2

$$2 \log \frac{||| \omega, h^k |||}{||| \omega_\varphi, h^k e^{-k\varphi} |||} = \sum_{j=0}^{\infty} k^{n+1-j} \int_0^1 dt \int_M \dot{\varphi}_t (\Delta_t a_{j-1}(g_t) - a_j(g_t)) \frac{\omega_t^n}{n!} \\ + \beta_{j-1}(g_\varphi) - \beta_{j-1}(g).$$

- By comparing coefficients of Method 1 and Method 2, we have for all $j \geq 0$,

$$\int_M iBC(\mathrm{Td}_j; g_\varphi, g) \frac{\omega^{n+1-j}}{(n+1-j)!} + \cdots \\ = \int_0^1 dt \int_M \dot{\varphi}_t (a_j(g_t) - \Delta_t a_{j-1}(g_t)) \frac{\omega_t^n}{n!} + \beta_{j-1}(g) - \beta_{j-1}(g_\varphi).$$

- Our moment maps will be derived from taking the derivative of this with respect to the Kähler potential φ .

- We begin with the expression from Method 1, involving Bott-Chern forms.
- After some integration by parts, we get

$$\frac{\delta}{\delta\varphi} \int_M iBC(\mathrm{Td}_j; g_\varphi, g) \frac{\omega^{n+1-j}}{(n+1-j)!} + \cdots = \frac{1}{(n-j)!} \int_M \delta\varphi \tilde{Z}_j(M, \omega) \omega^n$$

for some function $\tilde{Z}_j(M, \omega)$ depending on the curvature of the Kähler metric $g_{(\omega, J)}$.

- Interestingly, these $\tilde{Z}_j(M, \omega)$ are *Z-critical equations* introduced by Dervan-Hallam with central charge given by Todd class.

More on $\tilde{Z}$ -critical equations

- Dervan-Hallam gave the most general recipe to obtain scalar curvature equation for the Kähler metrics with moment map interpretation.
- That is, given invariant degree j polynomial ϕ , they define a $\tilde{Z}$ -critical equation $\tilde{Z}(\phi)$ so that

$$\int_M \tilde{Z}(\phi) \omega^n = \int_M \phi(TM, g) \omega^{n-j}$$

and has a moment map interpretation with respect to 2-form

$$\Omega_\phi = \int_{\mathcal{X}/B} \phi(TM_b, g|_{M_b}) \omega^{n+1-j}.$$

- They gave an explicit way to compute $\tilde{Z}(\phi)$, but one can compute them simply by taking derivative of the Bott-Chern forms corresponding to ϕ .

- From expression from Method 2, involving Bergman kernel and Ray-Singer torsion expansions, we get

$$\frac{n!}{(n-j)!} \tilde{Z}_j(M, \omega) = a_j(g) - \Delta a_{j-1}(g) + b_{j-1}(g), \quad (0.2)$$

for some functions b_j coming from analytic torsion.

- Now we will give a moment map interpretation of RHS of (0.2).

Derivation of the moment map equations

- Now we fix (M, ω) and vary J . Fix a prequantization $P = (L, h, \nabla)$.
- Let $\mathcal{G} = \text{Aut}(P)$ be the group of bundle diffeomorphisms of L that preserve h and ∇ . Then any element of $\mathcal{G}$ induces an element of $\text{Ham}(M, \omega)$ on the base M , and acts on $\mathcal{J}_{int}$ by pullback.
- Recall that over the universal family $\mathcal{X} = \cup_J M_J \rightarrow \mathcal{J}_{int}$, we have an hermitian holomorphic line bundle $(\tilde{L} = \cup_J L_J, \tilde{h})$.
- By determinant line bundle construction, we obtain a holomorphic line bundle $\lambda^{(k)} := \lambda(\tilde{L}^k)$ on $\mathcal{J}_{int}$ for each $k \in \mathbb{N}$, equipped with Quillen metric.
- It can be check that $\lambda^{(k)}$ is $\mathcal{G}$ -equivariant and Chern connection with respect to Quillen metric is $\mathcal{G}$ -invariant for each k .

Moment map from equivariant line bundle

- Since determinant line bundle and Quillen metric are not given in terms of local coordinates, following coordinate-independent characterization of moment map is convenient.
- Suppose there exists a G -equivariant complex line bundle λ on N endowed with a G -invariant connection ∇^λ whose curvature gives the G -invariant symplectic form Ω ($\nabla^2 = -i\Omega$).
- For each $X \in \mathfrak{g} = \text{Lie}(G)$, define

$$\tilde{\mu}(X) = \mathcal{L}^\lambda(X) - \nabla_{\hat{X}}^\lambda \in \Gamma(N, \text{End}\lambda) \simeq C^\infty(N, \mathbb{C}).$$

- Then the map $\mu : N \rightarrow \mathfrak{g}^*$ defined by $\langle \mu(\cdot), X \rangle = i\tilde{\mu}(X)$ is the moment map for $G \curvearrowright (N, \Omega)$.

- We apply this recipe to the determinant line bundle for each k and get a moment map

$$\tilde{\mu}^{(k)}(f) = \mathcal{L}^{\lambda^{(k)}}(f) - {}^1\nabla_{\hat{f}}^{\lambda^{(k)}} \in \Gamma(\mathcal{J}_{int}, \text{End} \lambda^{(k)}) \simeq C^\infty(\mathcal{J}_{int}, \mathbb{C})$$

for $f \in C^\infty(M) \simeq \text{Lie}(\mathcal{G})$.

- We can compute the asymptotic expansion of $\tilde{\mu}^{(k)}(f)$ as $k \rightarrow \infty$.
- The coefficients turn out to be

$$\mu_j(g_{(\omega, J)}) = a_j(g_{(\omega, J)}) - \Delta a_{j-1}(g_{(\omega, J)}) + b_{j-1}(g_{(\omega, J)}).$$

- Recall)

$$\frac{\delta}{\delta \varphi} \log || \cdot ||_{\omega_\varphi, h_\varphi^k}^2 = \sum_{j=0}^{n+1} k^{n+1-j} \int_M \delta \varphi \mu_j(g_{(\omega, J)}) \frac{\omega_\varphi^n}{n!}.$$

- The reason combination of TYZ coefficients of the form

$$a_j(g_{(\omega, J)}) - \Delta a_{j-1}(g_{(\omega, J)}) \quad (0.3)$$

appeared before (J fixed, φ varied) in μ_j was due to Donaldson's lemma, stating that

$$\delta \log |\cdot|_{\omega_\varphi, h_\varphi^k}^2 = \int_M \delta \varphi (\Delta_\varphi \rho_k(\omega_\varphi) - k \rho_k(\omega_\varphi)) \frac{\omega_\varphi^n}{n!}.$$

- Same combination (0.3) appears in the moment map situation (φ fixed, J varied) due to *Tuynman's lemma*:

$$\Pi(\nabla_{-X_f}^{L^k} + ikf) = \Pi(ikf - \Delta f)$$

where Π denotes the orthogonal projection onto the $H^0(M, L^k)$.

- $\mu^{(k)}$ are moment maps with respect to curvature 2-forms
 $\Omega^{(k)} = i(1 \nabla^{\lambda^{(k)}})^2$.
- By BGS curvature formula, we have

$$\begin{aligned}
 \Omega^{(k)} &= - \left[\int_{\mathcal{X}/\mathcal{J}_{int}} \text{Td}(TM, g_{(\omega, \mathcal{J})}) \text{ch}(\tilde{L}^k, \tilde{h}^k) \right]^{(2)} \\
 &= \sum_{j=0}^{n+1} k^{n+1-j} \left[- \int_{\mathcal{X}/\mathcal{J}_{int}} \text{Td}_j(TM, g_{(\omega, \mathcal{J})}) \text{ch}_{n+1-j}(\tilde{L}, \tilde{h}) \right] \\
 &=: \sum_{j=0}^{n+1} k^{n+1-j} [-\Omega_j]
 \end{aligned}$$

- Finally, by taking the asymptotic expansion of the moment map equation $d\langle \mu^{(k)}(\cdot), f \rangle = \iota_{\tilde{f}} \Omega^{(k)}$ and using the commutativity of differentiation over the base with asymptotic expansion, we deduce our main result.

Further remarks

- Note that the coefficients of the asymptotic expansion of $\log || \cdot ||_{\omega_\varphi, h_\varphi^k}^2$ give functionals on the space of Kähler potentials.
- For example, the k^{n+1} coefficient is the *Aubin-Yau functional*, and the k^n coefficient is the *Mabuchi functional*.
- In general, k^{n+1-j} -coefficient can be regarded as a functional whose critical point equation is $\mu_j \equiv 0$ (it can be easily modified to be $\mu_j \equiv \text{constant}$).
- By taking derivative in the direction of holomorphic vector field, we obtain a *higher Futaki invariants*, obstructing the existence of the critical points.

- That is, let X be a holomorphic vector field on M with zeros.
- There is a complex valued function θ_X so that

$$\iota_X \omega := \omega(X, \cdot) = -\bar{\partial} \theta_X.$$

- By taking derivative of those functionals with respect to the flow of $\operatorname{Re} X$, we get

$$\int_M i \operatorname{Im} \theta (\mu_j(\omega)) \frac{\omega^n}{n!} = \int_M \operatorname{Td}_j \left(R + \frac{\nabla X + (\bar{\nabla} X)^{b, \sharp}}{2} \right) \frac{(\omega + i \operatorname{Im} \theta)^{n+1-j}}{(n+1-j)!}.$$

- The right-hand side is a slight modification of the higher-Futaki invariant [Futaki '04] and it does not depend on ω . This formula is in fact conjectured by Z. Lu in 2004.

Thank you!